

A Nuclear-Laplace Reduction for the General Smallest-Singular-Value Problem

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Status

Conditional counterexample reduction, likely valid. The matrix construction, the exact Pfaffian formula, and the implication to a counterexample are proved. One scalar asymptotic dependency remains: an explicit Pfaffian ratio must be shown to be unbounded. Exact and high-precision computations support that dependency but are not a proof.

Source question

Livshyts, Fernandez, and Mui prove

$$\mathbb{P}\{s_n(A) \leq \varepsilon/\sqrt{n}\} \leq C\varepsilon + e^{-cn}$$

for square unconditional isotropic log-concave matrices, and ask whether the same behavior persists for general square isotropic log-concave matrix laws [1]. Their discussion on PDF page 4 is reproduced below.

Significant challenges arise in the implementation of this scheme. For instance, in the incompressible case, we need small ball estimates not for the isotropic log-concave vector (which is our matrix), but for a sub-vector (a column of our matrix) conditioned on the realization of the random normal (orthogonal to the span of the other columns of the matrix). Conditioning preserves log-concavity but loses isotropicity, and therefore we cannot use the slicing estimate [4, 21] ready-made. We manage to address this challenge in the tall case, the square unconditional case, as well as in the independent columns case. However, in general, this issue appears to be significant, and not just a minor technicality, as can be seen in some examples. In view of that, we are not convinced that the answer should remain the same in the general log-concave isotropic case; this question remains an interesting open problem.

Idea of the proof

The nuclear norm is the convex matrix analogue of the ℓ_1 norm, so the law with density $e^{-\|A\|_*}$ is an especially natural place for excessive mass near low rank. Orthogonal invariance makes its covariance scalar, hence isotropic normalization costs only one scalar.

For real square matrices the ordered singular values have density

$$e^{-\sum x_i} \prod_{i < j} (x_j^2 - x_i^2).$$

The density of the minimum at zero is therefore a ratio $d_n = B_n/Z_n$ of two partition functions. De Bruijn's identity turns both into completely explicit Pfaffians. More importantly, shifting all remaining singular values by x proves the quantitative inequality $g_n(x) \geq e^{-nx}d_n$. Thus an unbounded boundary ratio creates probability at least a diverging multiple of $1/n$ on a window whose isotropic parameter is at most $1/n$. This is incompatible with any universal linear small-ball constant.

The ensemble and the conditional result

Let X_n be a random real $n \times n$ matrix with density

$$d\mu_n(A) = \mathcal{Z}_n^{-1} e^{-\|A\|_*} dA. \quad (1)$$

The density is log-concave and invariant under $A \mapsto UAV$ for $U, V \in O(n)$. It is centered, and its entry covariance is $v_n I_{n^2}$ for some $v_n > 0$. Put $\tilde{X}_n = X_n/\sqrt{v_n}$.

For $0 < x_1 < \dots < x_n$, define

$$Z_n = \int e^{-\sum_{i=1}^n x_i} \prod_{1 \leq i < j \leq n} (x_j^2 - x_i^2) dx_1 \dots dx_n, \quad (2)$$

$$B_n = \int_{0 < y_1 < \dots < y_{n-1}} e^{-\sum_{j=1}^{n-1} y_j} \prod_{j=1}^{n-1} y_j^2 \prod_{1 \leq i < j \leq n-1} (y_j^2 - y_i^2) dy. \quad (3)$$

For $n = 1$, the empty integral defining B_1 is 1. Write $d_n = B_n/Z_n$.

Theorem 1 (conditional counterexample). *If d_n is unbounded along a sequence of dimensions, then no absolute constants $C, c > 0$ can satisfy*

$$\mathbb{P}\{s_n(A) \leq \varepsilon/\sqrt{n}\} \leq C\varepsilon + e^{-cn}$$

for every square isotropic log-concave random matrix A and every $\varepsilon > 0$. The isotropic matrices $\tilde{X}_n$ from (1) give the counterexample sequence.

Proof of the conditional theorem

The real square SVD change of variables shows that the ordered singular values of X_n have joint density Z_n^{-1} times the integrand in (2); the orthogonal-group constant cancels in the normalization. Let g_n be the density of the least singular value. Directly,

$$g_n(x) = \frac{e^{-x}}{Z_n} \int_{x < y_1 < \dots < y_{n-1}} e^{-\sum y_j} \prod_j (y_j^2 - x^2) \prod_{i < j} (y_j^2 - y_i^2) dy. \quad (4)$$

In particular, $g_n(0) = B_n/Z_n = d_n$.

Lemma 1 (a quantitative boundary window). *For every $x \geq 0$,*

$$g_n(x) \geq e^{-nx}d_n.$$

Proof. In (4), substitute $y_j = z_j + x$. The exponential contributes e^{-nx} . For $i < j$,

$$(z_j + x)^2 - (z_i + x)^2 = (z_j - z_i)(z_j + z_i + 2x) \geq z_j^2 - z_i^2,$$

and

$$(z_j + x)^2 - x^2 = z_j(z_j + 2x) \geq z_j^2.$$

The shifted integral is therefore at least B_n . $\square$

It remains to track isotropic normalization. Orthogonal row and column sign flips kill every covariance between distinct entries, while row and column permutations make all entry variances equal. Thus $\text{Cov}(\text{vec } X_n) = v_n I$.

There is also a useful dimension-free lower bound on v_n . Radial integration for any norm in dimension $D = n^2$ gives

$$\mathbb{E}\|X_n\|_*^2 = D(D+1) = n^2(n^2+1).$$

Since $\|A\|_* \leq \sqrt{n}\|A\|_F$,

$$n^2 v_n = \mathbb{E}\|X_n\|_F^2 \geq \frac{1}{n} \mathbb{E}\|X_n\|_*^2, \quad v_n \geq n + \frac{1}{n}.$$

Integrating Lemma 1 on $[0, 1/n]$ yields

$$\mathbb{P}\{s_n(X_n) \leq 1/n\} \geq \frac{1 - e^{-1}}{n} d_n. \quad (5)$$

For

$$\varepsilon_n = \frac{1}{\sqrt{n}\sqrt{v_n}},$$

the event in (5) is exactly $\{s_n(\tilde{X}_n) \leq \varepsilon_n/\sqrt{n}\}$, and the preceding variance estimate gives $\varepsilon_n \leq 1/n$. If universal C, c existed, (5) would imply

$$(1 - e^{-1})d_n \leq C + ne^{-cn}.$$

This contradicts unboundedness of d_n and proves Theorem 1. $\square$

Exact Pfaffian reduction

The remaining dependency is not a high-dimensional integral. It is an explicit Pfaffian ratio. For integers $0 \leq i < j$, set

$$q_{ij} = \frac{1}{(2i)!(2j)!} \int_{0 < x < y} e^{-x-y} (x^{2i} y^{2j} - x^{2j} y^{2i}) dx dy \quad (6)$$

$$= 2I_{1/2}(2i+1, 2j+1) - 1, \quad (7)$$

and put $q_{ji} = -q_{ij}$, $q_{ii} = 0$. For an ordered finite index set S , let $\text{DPf}(S)$ denote the Pfaffian of $(q_{ij})_{i,j \in S}$ when $|S|$ is even. When $|S|$ is odd, augment this skew matrix by a last column of ones and a last row of minus ones before taking the Pfaffian.

Proposition 1 (de Bruijn formula). *With the sign convention in (6),*

$$d_n = \frac{\text{DPf}(\{1, \dots, n-1\})}{\text{DPf}(\{0, \dots, n-1\})}.$$

Proof. Write the Vandermonde factor as

$$\prod_{i < j} (x_j^2 - x_i^2) = \det[x_j^{2(i-1)}]_{i,j=1}^n.$$

De Bruijn’s ordered-chamber integration identity expresses its integral against $\prod e^{-x_j} dx_j$ as the Pfaffian of the skew moments in (6); for odd size the unpaired moments are $\int_0^\infty x^{2i} e^{-x} dx = (2i)!$, producing the one-vector after normalization. The factors $\prod_{i=0}^{n-1} (2i)!$ occur in Z_n . For B_n , the basis exponents are $2, 4, \dots, 2(n-1)$, so the factors are $\prod_{i=1}^{n-1} (2i)!$. Their ratio is $1/0! = 1$, proving the formula. Finally, (7) follows by interpreting normalized gamma integrals: if G_i, G_j are independent gamma variables of shapes $2i+1, 2j+1$, then $G_i/(G_i + G_j)$ is beta distributed. $\square$

Conditional dependencies and computation

The only unproved statement is

$$\sup_n d_n = \infty. \tag{8}$$

No rate is needed. The supplied exact-fraction program implements Proposition 1 and gives:

n	1	10	40	100	200
d_n	1.000000	2.038100	2.577737	2.917386	3.168924

Values through $n = 40$ were computed exactly as rational numbers; a separate high-precision implementation continued the audit through $n = 200$. The data are consistent with growth comparable to $\sqrt{\log n}$, but neither that fit nor the weaker dependency (8) is claimed as proved.

Limitations and novelty check

This packet is conditional, not a claimed disproof. A bounded search used the exact source title and arXiv id together with “smallest singular value Schatten 1 ball”, “nuclear norm random matrix hard edge”, and close Pfaffian/partition-function phrases. The closest inspected sources concerned macroscopic Schatten-ball equilibrium measures [2, 3], the critical Freud ensemble [4], and hard-edge universality for other Laguerre-type weights [5]. No source was found that states (8) or this counterexample reduction.

Human-review recommendation

The matrix argument should be checked by a specialist in asymptotic convex geometry; the remaining ratio should be sent to a specialist in Pfaffian ensembles or skew-orthogonal polynomials. Proving any divergence of the ratio would immediately promote this packet to a full counterexample.

References

- [1] G. V. Livshyts, M. Fernandez V, and S. Mui, *On the Smallest Singular Value of Log-Concave Random Matrices*, arXiv:2508.17745 (2025).
- [2] B. Dadoun, M. Fradelizi, O. Guédon, and P.-A. Zitt, *Asymptotics of the inertia moments and the variance conjecture in Schatten balls*, J. Funct. Anal. 284 (2023), 109752; arXiv:2111.07803.

- [3] Z. Kabluchko, J. Prochno, and C. Thäle, *Exact asymptotic volume and volume ratio of Schatten unit balls*, J. Approx. Theory 257 (2020), 105457; arXiv:1804.03467.
- [4] T. Claeys, I. Krasovsky, and O. Minakov, *Weak and strong confinement in the Freud random matrix ensemble and gap probabilities*, Comm. Math. Phys. 402 (2023), 285–340; arXiv:2209.07253.
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